1991AJ....102.1753D

THE ASTRONOMICAL JOURNAL VOLUME 102, NUMBER 5 NOVEMBER 1991

NEAR-IR OBSERVATIONS OF 101 Be STARS

S. M. DOUGHERTY, A. R. TAYLOR, AND T. A. CLARK
Department of Physics and Astronomy, 2500 University Dr. NW., Calgary T2N 1N4, Alberta, Canada
Received 17 May 1991; revised 18 July 1991

ABSTRACT

Observations of 101 Be stars taken over a two year period in the near-IR between 1 and 5 μm are presented and discussed. The near-IR color excess of all program stars is derived, and found to increase with wavelength for all these stars. The fraction of stars with color excess doubles between 1.25 and 3.6 μm. There appears to be an upper limit to the magnitude of the color excess as a function of stellar type, with early type stars having a higher upper limit than later spectral types. No correlation of the presence or magnitude of color excess with projected rotational velocity is evident. We calculate the spectral index of the excess emission spectra for stars having color excess. On average the spectral index through the near- and far-IR *IRAS* wavelength regimes is constant. There is evidence that some stars have excess emission with spectral index values outside the range expected for free–free and bound–free emission. This is attributed to either dust emission or the effect of absorption of photospheric emission by cool circumstellar material along the line of sight to the star.

1. INTRODUCTION

Be stars are nonsupergiant stars that show, or have shown at some time, hydrogen emission lines. The emission lines are known to be often highly variable. Their continuum spectra are characterized by an excess of emission at wavelengths greater than ~ 1 μm compared to the spectra of "normal" B-type absorption line stars. The hydrogen line emission is due to the recombination of ionized hydrogen in an extended circumstellar envelope surrounding the central star. The IR excess arises from both bound–free and free–free emission from proton-electron scattering in the circumstellar material (e.g., see Woolf *et al.* 1970; Gehrz *et al.* 1974; Poeckert & Marlborough 1978) within 20–60$R_\odot$ of the central star (Waters 1986). Radio and millimeter observations (Taylor *et al.* 1987, 1990; Waters *et al.* 1989; 1991) show that the circumstellar material sometimes extends to large distances from the central star ($\sim 1000R_\odot$ in some cases). Comparison of the far-IR–radio spectral index to the slope of the far-IR energy distribution indicates a dramatic change in the structure of the circumstellar envelope in the outer regions of several Be star envelopes (Taylor *et al.* 1987; Waters *et al.* 1991). Variations in the structure of the outer regions of the circumstellar envelope over a period of a year are evident in recent multi-epoch radio observations (Dougherty *et al.* 1991).

Further understanding of the structure of the envelope should reside in the manner in which variations in the circumstellar emission at different wavelength regions can be correlated. Emission line and continuum variations at optical wavelengths have been studied extensively (e.g., see Doazan 1982; Slettebak & Snow 1987; Baade 1991) but variations at IR wavelengths have been less well studied. In a survey of 20 Be stars between 1 and 20 μm, Gehrz *et al.* (1974) found no evidence for variability on a timescale of a year. More recently, Dachs & Wamsteker (1982) found evidence for near-IR variations in 10 stars from a sample of 46 southern Be stars over a six year epoch. They also report a correlation between IR variations and variations in the equivalent width of the Balmer emission lines, supporting the hypothesis that IR and optical variations have a common origin attributed to emission from a circumstellar envelope of varying density. Ashok *et al.* (1984) also found evidence of IR variations in four stars from a survey of 35 Be stars.

The incidence of IR variability in each of the surveys mentioned above appears to be very low compared to the number of Be stars that show variations at optical wavelengths. This is somewhat surprising, considering the correlation of variations in Hα equivalent width and IR variations reported by Dachs and Wamsteker. A larger database of IR observations would allow a more thorough investigation of the incidence of IR variability in Be stars. However, there has been no recent survey of a large sample of Be stars over a wide range of epochs. Moreover, most past work has been out to only 2.2 μm. To address this issue, we have carried out a near-IR survey of 101 Be stars at several epochs over a two year period out to 3.6 μm. In this paper (Paper I) we present the observations from the survey. Multi-epoch observations are averaged and near-IR excesses calculated and the properties of the IR excess are discussed. The identification and analysis of Be stars with variable IR continuum emission will be described in a second paper (Paper II).

2. OBSERVATIONS

2.1 Program Stars

The stars in the survey were selected from all Be stars in the Bright Star Catalogue (BSC) (Hofleit & Jaschek 1982) with a declination greater than $-30°$. Several of the Be stars in the list of Slettebak (1982) that are listed as B stars in the BSC were also included. The southern declination limit was imposed by the most southern observatory used in the survey, the University of Hawaii Mauna Kea Observatory (MKO).

2.2 The Photometry

The program stars were observed over several epochs between April 1988 and March 1991 at three observatories: the University of Calgary Rothney Astrophysical Observatory (RAO) 1.5 m telescope, the National Optical Astron-

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

omy Observatory,[1] Kitt Peak 1.3 m telescope and the MKO 0.6 m telescope. At all sites an InSb detector, cooled by solid nitrogen, with a 15″–30″ aperture and standard broadband *JHKLM* filters, was used. Observations were made using the standard procedure of chopping between a source + sky position and a sky position. The separation between the source and sky positions was ~60″ and the chopping frequency was 10 Hz.

A total of 101 program stars were observed at *J*, *H*, and *K* passbands out of 111 Be stars in the BSC with declination greater than −30°. At *L* and *M* we observed 72 and 30 stars, respectively. This is a reasonably complete survey of bright Be stars considering that the BSC contains 165 Be stars. Due to bad weather and local sidereal time restrictions, 39 of the survey stars were observed at only one epoch. The remaining stars were observed during at least two observing sessions. In addition to the program stars, observations of between five and ten standard stars were obtained per observing run. Atmospheric extinction corrections were calculated for each observing night from several observations of a standard star at several different altitudes. Secondary standards, ranging in spectral type from B1 to M2 were also observed for calculation of color corrections. The IR magnitudes of these standard stars were taken from Koorneef (1983a) and the data from each site were transformed into one homogenous dataset.

The IR magnitudes of the survey stars are presented in Table 1. In the first three columns the HR number, star name, and Julian Date of the observations are listed. The *JHKLM* photometry is given in columns 4–8. The quoted errors of each observation include the uncertainty of the transformation to the standard system and the error due to variations in the sky transparency during each observing run. These errors are within the range of those quoted by Dachs & Wamsteker (1982).

3. NEAR-INFRARED EXCESSES

The weighted averages of the multi-epoch observations were calculated to give an average magnitude at each passband over the period of the observations. These are quoted in Table 2. Where more than one observation of a star was made, the number of observations in each average is included in brackets after each entry.

Before calculating the IR excesses, the magnitudes in Table 2 were dereddened to account for the effects of interstellar extinction. Values of $E(B-V)$ were calculated from $(B-V)$ colors taken from the *BSC* and the intrinsic $(B-V)_*$ colors for "normal" main sequence B stars based on spectral type (Fitzgerald 1970). The near-IR extinction law from Koorneef (1983b) was adopted and the observed colors $(V-m_\lambda)_{\rm obs}$ were corrected to yield dereddened $(V-m_\lambda)_0$ color values, where m_λ is the observed magnitude at wavelength λ and V is the V band magnitude taken from the BSC. The color excess $CE(V-m_\lambda)$ was then calculated from

$$CE(V-m_\lambda) = (V-m_\lambda)_0 - (V-m_\lambda)_*, \quad (1)$$

where $(V-m_\lambda)_*$ is the intrinsic color of "normal" main sequence B stars, taken from Koorneef (1983b).

The calculated values of color excess may be influenced by several factors which we have not included in our calculations. Early type Be stars are known to have flat energy distributions in the Paschen continuum because of an excess emission due to ionized circumstellar material. In this case, $(B-V)$ would be larger than expected, giving an overestimate of $E(B-V)$. Schild (1978) estimates this effect to be as large as 0.15^m for B0–B3 stars, but negligible for later types. Hence the IR interstellar reddening correction could be overestimated, which would result in the color excess as defined in Eq. (1) being *underestimated* for these spectral types by up to 0.2^m at *J* band, and 0.3^m at *L*. Furthermore, we have not accounted for either the uncertainty in the spectral type classification or the intrinsic $(B-V)$ colors of normal stars. The effect of misclassification of spectral type will be greatest for later spectral types where the intrinsic color changes by as much as 0.1^m between B8 and B9. The size of these effects is significant compared to the errors on the colors and should be kept in mind during the following discussion.

4. THE EXCESS SPECTRUM

The energy distribution of Be stars consists of a contribution from the photosphere of the underlying B-star and a contribution from the circumstellar material that gives rise to an excess of emission. If we denote the dereddened observed flux and the photospheric flux at wavelength λ as f_λ^0 and f_λ^*, respectively, then the excess flux, f_λ^e, due to circumstellar material is

$$f_\lambda^e = f_\lambda^0 - f_\lambda^*. \quad (2)$$

Dividing by the photospheric flux gives

$$f_\lambda^e / f_\lambda^* = Z_\lambda - 1, \quad (3)$$

where $Z_\lambda = f_\lambda^0 / f_\lambda^*$ is the so-called monochromatic flux ratio (Lamers & Waters 1984). If the spectra of the photospheric and circumstellar emission can be represented by a power law of the form $f \propto \lambda^{-\alpha}$ over some wavelength range $\Delta\lambda = \lambda_2 - \lambda_1$ $(\lambda_2 > \lambda_1)$, where α is the spectral index, then Eqs. (2) and (3) lead to the expression

$$(\alpha_e - \alpha_*)\log(\lambda_1/\lambda_2) = \log\left(\frac{Z_{\lambda_2} - 1}{Z_{\lambda_1} - 1}\right), \quad (4)$$

where α_e is the spectral index of the excess emission spectrum and α_* is the spectral index of the photospheric emission between wavelengths λ_1 and λ_2. We can calculate α_* from intrinsic colors since

$$\alpha_* \log(\lambda_1/\lambda_2) = 0.4(m_{\lambda_1}^* - m_{\lambda_2}^*) + \text{constant}, \quad (5)$$

where the constant is determined from absolute flux calibrations at wavelengths λ_1 and λ_2.

Z_λ can be determined from the excess colors in the following manner. The color excess $CE(V-m_\lambda)$, defined in Eq. (1), may be written in the form

$$CE(V-m_\lambda) = (V^0 - V^*) - (m_\lambda^0 - m_\lambda^*) \quad (6)$$

$$= 2.5\log(Z_\lambda/Z_V). \quad (7)$$

Substituting Eqs. (7) and (5) into Eq. (4) then gives

$$(\alpha_e - \alpha_*)\log(\lambda_1/\lambda_2) = \log\left(\frac{Z_V 10^a - 1}{Z_V 10^b - 1}\right), \quad (8)$$

where $a = 0.4\,CE(V-m_{\lambda_2})$ and $b = 0.4\,CE(V-m_{\lambda_1})$.

If we know the value of Z_V for each of the program stars

[1] The National Optical Astronomy Observatories are operated by the Association of Universities for Research in Astronomy, Inc., under cooperative agreement with the National Science Foundation

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 1. Multi-epoch near-IR observations of Be stars.

HR	Star Name	Julian Date[a]	J	H	K	L	M
7	10 Cas	2447817.97	5.57 ± 0.03	5.59 ± 0.03	5.57 ± 0.03	–	–
		8070.12	5.59 ± 0.03	5.53 ± 0.03	5.52 ± 0.03	–	–
193	*o* Cas	7798.87	4.37 ± 0.03	4.27 ± 0.04	4.08 ± 0.04	3.71 ± 0.16	–
		8070.07	4.35 ± 0.03	4.27 ± 0.04	4.10 ± 0.04	3.87 ± 0.07	–
264	γ Cas	7879.81	2.19 ± 0.04	2.08 ± 0.05	1.88 ± 0.05	1.52 ± 0.06	1.24 ± 0.24
		8067.13	2.15 ± 0.03	2.05 ± 0.04	1.86 ± 0.04	1.48 ± 0.06	–
335	ϕ And	7798.83	4.42 ± 0.03	4.38 ± 0.03	4.37 ± 0.04	–	–
		7879.85	4.32 ± 0.03	4.36 ± 0.04	4.32 ± 0.05	4.24 ± 0.05	–
496	ϕ Per	7877.77	3.93 ± 0.04	3.82 ± 0.05	3.55 ± 0.05	3.09 ± 0.07	–
894		7817.99	6.24 ± 0.03	6.24 ± 0.04	6.19 ± 0.04	–	–
985	BK Cam	7806.93	4.92 ± 0.03	4.88 ± 0.03	4.71 ± 0.03	4.42 ± 0.06	–
		7879.89	4.95 ± 0.04	4.88 ± 0.05	4.74 ± 0.05	4.39 ± 0.06	–
1087	ψ Per	7795.01	4.42 ± 0.03	4.38 ± 0.03	4.24 ± 0.03	3.91 ± 0.05	–
		7877.71	4.28 ± 0.04	4.21 ± 0.06	4.08 ± 0.05	3.74 ± 0.07	–
1126	13 Tau	7803.79	5.90 ± 0.03	5.90 ± 0.03	5.89 ± 0.03	6.11 ± 0.28	–
		7875.83	5.70 ± 0.03	5.73 ± 0.04	5.72 ± 0.04	5.59 ± 0.06	–
1142	17 Tau	7797.93	4.05 ± 0.03	4.09 ± 0.03	4.11 ± 0.03	–	–
		7875.71	3.87 ± 0.03	3.93 ± 0.04	3.90 ± 0.04	3.91 ± 0.04	3.99 ± 0.30
1156	23 Tau	7796.96	4.32 ± 0.03	4.26 ± 0.03	4.16 ± 0.03	–	–
		7873.76	4.16 ± 0.03	4.15 ± 0.04	4.03 ± 0.04	3.77 ± 0.05	3.42 ± 0.19
1160		8246.76	6.00 ± 0.04	5.94 ± 0.06	6.05 ± 0.07	–	–
1165	η Tau	7794.97	3.08 ± 0.03	3.12 ± 0.03	3.08 ± 0.03	2.96 ± 0.05	–
		7797.91	3.16 ± 0.03	3.16 ± 0.03	3.14 ± 0.03	2.97 ± 0.17	–
		7873.73	2.95 ± 0.04	3.00 ± 0.05	3.01 ± 0.05	2.97 ± 0.05	3.06 ± 0.21
1180	28 Tau	7797.98	5.20 ± 0.03	5.21 ± 0.04	5.07 ± 0.04	–	–
		7875.77	5.12 ± 0.03	5.08 ± 0.03	5.02 ± 0.04	4.68 ± 0.05	3.83 ± 0.24
1273	48 Per	7807.03	3.99 ± 0.03	3.95 ± 0.03	3.81 ± 0.03	3.61 ± 0.06	–
		7877.74	3.91 ± 0.04	3.90 ± 0.06	3.78 ± 0.05	3.50 ± 0.07	–
1500		7818.04	5.90 ± 0.03	5.88 ± 0.03	5.85 ± 0.04	–	–
		7879.97	5.90 ± 0.03	5.84 ± 0.05	5.75 ± 0.05	5.59 ± 0.07	–
1622	11 Cam	7880.04	5.22 ± 0.04	5.13 ± 0.05	4.99 ± 0.05	4.59 ± 0.06	–
1660	105 Tau	7803.89	5.44 ± 0.03	5.27 ± 0.04	5.02 ± 0.04	4.82 ± 0.27	–
		7879.92	5.53 ± 0.04	5.35 ± 0.05	5.17 ± 0.05	4.76 ± 0.06	–
1789	ψ Ori	7875.93	5.13 ± 0.03	5.22 ± 0.03	5.27 ± 0.04	5.28 ± 0.05	–
1858	120 Tau	7803.88	5.85 ± 0.03	5.78 ± 0.03	5.69 ± 0.04	–	–
		7875.86	5.63 ± 0.03	5.60 ± 0.04	5.50 ± 0.04	5.21 ± 0.08	–
1910	ζ Tau	7803.83	3.41 ± 0.03	3.35 ± 0.03	3.18 ± 0.04	2.89 ± 0.06	–
		7877.78	3.14 ± 0.04	3.11 ± 0.06	2.99 ± 0.05	2.67 ± 0.07	–
1934	ω Ori	7807.00	4.52 ± 0.03	4.43 ± 0.03	4.33 ± 0.04	4.13 ± 0.27	–
		7873.87	4.44 ± 0.04	4.35 ± 0.04	4.16 ± 0.04	3.72 ± 0.06	3.60 ± 0.22
2309		7876.06	6.28 ± 0.03	6.35 ± 0.04	6.34 ± 0.04	6.28 ± 0.11	–
2343	ν Gem	7880.09	4.40 ± 0.03	4.45 ± 0.04	4.43 ± 0.05	4.37 ± 0.05	–
2356	β Mon	7945.70	4.29 ± 0.04	4.14 ± 0.05	3.98 ± 0.05	3.90 ± 0.14	–
2568	ψ^9 Aur	7880.06	6.02 ± 0.03	6.03 ± 0.05	5.97 ± 0.05	5.82 ± 0.07	–
2825	FW CMa	7875.99	5.25 ± 0.03	5.19 ± 0.04	5.04 ± 0.04	4.71 ± 0.05	–
2845	β CMi	7639.70	3.02 ± 0.03	3.09 ± 0.04	3.03 ± 0.04	2.96 ± 0.04	–
		7875.96	3.06 ± 0.03	3.08 ± 0.03	3.07 ± 0.03	2.99 ± 0.04	2.82 ± 0.15
		7945.75	3.09 ± 0.03	3.10 ± 0.04	3.09 ± 0.04	3.14 ± 0.07	–
3135		7640.65	6.49 ± 0.04	6.39 ± 0.05	6.19 ± 0.05	5.82 ± 0.06	5.72 ± 0.19
3858		7641.67	5.17 ± 0.04	5.06 ± 0.05	4.89 ± 0.05	4.69 ± 0.05	4.35 ± 0.12
3946		7641.70	6.18 ± 0.04	6.10 ± 0.05	5.93 ± 0.05	5.68 ± 0.05	5.54 ± 0.20
4123		7640.68	5.54 ± 0.03	5.52 ± 0.05	5.51 ± 0.05	5.46 ± 0.05	5.48 ± 0.18
		8057.79	5.70 ± 0.04	5.72 ± 0.05	5.62 ± 0.05	5.70 ± 0.07	–
4696	ζ Crv	7640.71	5.37 ± 0.04	5.39 ± 0.05	5.39 ± 0.05	5.32 ± 0.05	5.60 ± 0.24
		8057.82	5.50 ± 0.03	5.53 ± 0.04	5.48 ± 0.05	5.46 ± 0.07	–
4787	κ Dra	7639.86	4.03 ± 0.03	4.02 ± 0.04	3.89 ± 0.04	3.59 ± 0.04	2.90 ± 0.25
		7640.73	3.95 ± 0.03	3.89 ± 0.04	3.80 ± 0.05	3.50 ± 0.05	3.21 ± 0.16
		7641.76	4.12 ± 0.03	4.11 ± 0.04	3.99 ± 0.04	3.71 ± 0.04	3.40 ± 0.09
		8058.75	3.91 ± 0.04	3.92 ± 0.07	3.83 ± 0.06	3.18 ± 0.13	–

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 1. (continued)

HR	Star Name	Julian Date[a]	J	H	K	L	M
5250	47 Hya	8059.75	5.39 ± 0.03	5.47 ± 0.03	5.45 ± 0.03	5.50 ± 0.08	–
5778	θ CrB	7640.77	4.44 ± 0.03	4.47 ± 0.05	4.44 ± 0.05	4.44 ± 0.05	4.52 ± 0.16
		8056.84	4.40 ± 0.03	4.49 ± 0.03	4.51 ± 0.04	4.39 ± 0.05	–
5907		8068.79	5.46 ± 0.03	5.46 ± 0.03	5.45 ± 0.03	5.45 ± 0.07	–
5938	4 Her	7639.77	5.91 ± 0.03	5.94 ± 0.04	5.97 ± 0.03	5.95 ± 0.04	5.75 ± 0.25
		7640.76	5.94 ± 0.03	5.94 ± 0.05	5.99 ± 0.05	5.98 ± 0.05	6.13 ± 0.23
		7641.77	5.99 ± 0.03	6.07 ± 0.04	6.08 ± 0.04	6.11 ± 0.04	5.96 ± 0.13
		8056.92	6.04 ± 0.03	6.05 ± 0.04	6.03 ± 0.04	–	–
		8068.77	5.89 ± 0.03	5.92 ± 0.03	5.93 ± 0.03	6.02 ± 0.16	–
5941	48 Lib	7641.82	4.90 ± 0.03	4.85 ± 0.04	4.71 ± 0.05	4.65 ± 0.05	4.30 ± 0.10
		8058.84	4.94 ± 0.04	4.87 ± 0.06	4.72 ± 0.05	4.23 ± 0.12	–
6118	χ Oph	7641.84	3.43 ± 0.04	3.21 ± 0.05	2.93 ± 0.05	2.51 ± 0.05	2.22 ± 0.11
		8056.99	3.39 ± 0.04	3.17 ± 0.04	2.93 ± 0.05	2.57 ± 0.07	2.20 ± 0.17
6422		8058.87	5.90 ± 0.04	5.89 ± 0.08	5.86 ± 0.06	–	–
		8068.85	6.00 ± 0.03	5.95 ± 0.03	5.91 ± 0.04	–	–
6519	51 Oph	8058.92	4.64 ± 0.04	4.59 ± 0.07	4.32 ± 0.06	3.16 ± 0.14	–
6621	V3894 Sgr	8068.87	5.92 ± 0.03	5.85 ± 0.04	5.71 ± 0.04	–	–
6664	88 Her	8069.89	6.97 ± 0.03	6.99 ± 0.04	7.30 ± 0.11	–	–
6712	66 Oph	7639.80	4.90 ± 0.03	4.88 ± 0.05	4.58 ± 0.04	4.14 ± 0.05	–
		7641.90	4.85 ± 0.03	4.79 ± 0.04	4.56 ± 0.04	4.17 ± 0.05	3.73 ± 0.10
		8056.98	4.91 ± 0.03	4.90 ± 0.03	4.75 ± 0.04	4.11 ± 0.05	–
6720		7639.82	6.47 ± 0.03	6.48 ± 0.04	6.40 ± 0.04	6.23 ± 0.04	5.62 ± 0.29
		8058.96	6.41 ± 0.04	6.42 ± 0.06	6.39 ± 0.05	–	–
6873		7639.83	6.24 ± 0.03	6.30 ± 0.04	6.27 ± 0.04	6.25 ± 0.04	–
		8059.90	6.23 ± 0.03	6.28 ± 0.03	6.23 ± 0.03	–	–
6881		8068.91	5.65 ± 0.03	5.90 ± 0.09	5.65 ± 0.04	–	–
6929	V4031 Sgr	8066.98	6.44 ± 0.03	6.47 ± 0.04	6.46 ± 0.04	–	–
		8069.01	6.51 ± 0.04	6.46 ± 0.05	6.46 ± 0.05	–	–
6971		7806.74	6.64 ± 0.03	6.64 ± 0.04	6.60 ± 0.04	–	–
		8068.90	6.69 ± 0.03	6.72 ± 0.03	6.68 ± 0.04	–	–
6984		7806.77	6.33 ± 0.03	6.37 ± 0.04	6.36 ± 0.04	6.41 ± 0.25	–
		8059.92	6.36 ± 0.03	6.42 ± 0.03	6.38 ± 0.03	–	–
7040	4 Aql	8069.90	5.11 ± 0.03	5.15 ± 0.03	5.12 ± 0.03	4.96 ± 0.06	–
7084	CX Dra	7639.87	5.79 ± 0.03	5.64 ± 0.04	5.55 ± 0.04	5.42 ± 0.04	4.89 ± 0.25
		7640.80	5.92 ± 0.04	5.77 ± 0.05	5.68 ± 0.05	5.61 ± 0.06	5.43 ± 0.20
		7641.82	5.80 ± 0.04	5.60 ± 0.05	5.50 ± 0.04	5.30 ± 0.05	5.27 ± 0.12
		8059.86	5.75 ± 0.03	5.61 ± 0.03	5.50 ± 0.03	5.45 ± 0.23	–
		8191.63	5.91 ± 0.04	5.77 ± 0.04	5.73 ± 0.07	–	–
7158	64 Ser	8066.92	5.47 ± 0.03	5.47 ± 0.04	5.46 ± 0.04	5.50 ± 0.08	–
7249	V4024 Sgr	7641.98	5.31 ± 0.04	5.36 ± 0.05	5.25 ± 0.06	5.06 ± 0.05	4.86 ± 0.12
		8059.01	5.67 ± 0.04	5.72 ± 0.07	5.66 ± 0.05	5.44 ± 0.12	–
7335		8059.95	7.04 ± 0.03	7.18 ± 0.04	7.22 ± 0.04	–	–
7403		7641.91	6.34 ± 0.03	6.32 ± 0.04	6.13 ± 0.04	5.85 ± 0.04	5.86 ± 0.16
		8058.11	6.37 ± 0.04	6.34 ± 0.05	6.19 ± 0.05	5.79 ± 0.07	–
7415	V923 Aql	8069.92	5.98 ± 0.03	5.96 ± 0.03	5.92 ± 0.03	–	–
		8191.67	5.65 ± 0.05	5.59 ± 0.06	5.65 ± 0.07	–	–
7418	6 β^2 Cyg	8068.97	5.24 ± 0.03	5.24 ± 0.03	5.23 ± 0.03	5.15 ± 0.07	–
7457	11 Cyg	7817.80	6.05 ± 0.03	6.09 ± 0.04	6.05 ± 0.04	–	–
		8057.05	6.19 ± 0.03	6.22 ± 0.03	6.21 ± 0.04	5.73 ± 0.22	–
7554	V1339 Aql	8066.96	6.51 ± 0.03	6.50 ± 0.04	6.34 ± 0.05	–	–
7565	12 Vul	7641.92	4.97 ± 0.03	4.99 ± 0.04	4.94 ± 0.04	4.81 ± 0.04	4.76 ± 0.11
		8057.12	5.05 ± 0.03	5.08 ± 0.03	5.03 ± 0.04	4.52 ± 0.11	–
7647	25 Cyg	7641.88	5.36 ± 0.03	5.40 ± 0.04	5.31 ± 0.04	5.15 ± 0.04	5.02 ± 0.15
		7803.55	5.32 ± 0.03	5.35 ± 0.03	5.28 ± 0.03	5.17 ± 0.27	–
		8057.08	5.38 ± 0.03	5.43 ± 0.03	5.35 ± 0.04	4.96 ± 0.14	–
7708	28 Cyg	7641.89	5.14 ± 0.03	5.14 ± 0.04	5.07 ± 0.04	4.88 ± 0.04	4.81 ± 0.11
		7803.59	5.15 ± 0.03	5.16 ± 0.03	5.03 ± 0.03	–	–
		8059.08	5.27 ± 0.04	5.30 ± 0.06	5.23 ± 0.05	5.03 ± 0.11	–
7719	20 Vul	7641.93	6.06 ± 0.03	6.10 ± 0.04	6.07 ± 0.04	5.97 ± 0.04	5.93 ± 0.17
		8059.99	6.09 ± 0.03	6.12 ± 0.03	6.09 ± 0.03	–	–

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 1. (continued)

HR	Star Name	Julian Date[a]	J	H	K	L	M
7739		8060.00	5.18 ± 0.03	5.26 ± 0.03	5.25 ± 0.03	5.21 ± 0.06	–
7789	25 Vul	7641.99	5.64 ± 0.03	5.72 ± 0.04	5.71 ± 0.04	5.56 ± 0.04	5.54 ± 0.13
		8060.03	5.65 ± 0.03	5.70 ± 0.03	5.64 ± 0.03	5.59 ± 0.07	–
7803		8069.95	6.19 ± 0.03	6.22 ± 0.05	6.15 ± 0.04	–	–
7807		7641.97	6.21 ± 0.03	6.20 ± 0.04	6.02 ± 0.04	5.70 ± 0.04	5.44 ± 0.14
		8067.02	6.26 ± 0.03	6.25 ± 0.04	6.16 ± 0.04	5.82 ± 0.08	–
7836	1 Del	8070.06	6.06 ± 0.03	6.08 ± 0.03	6.18 ± 0.06	–	–
7843		8069.00	6.00 ± 0.03	5.96 ± 0.04	5.93 ± 0.04	–	–
7890		8066.96	6.38 ± 0.03	6.40 ± 0.04	6.39 ± 0.04	–	–
7927	V568 Cyg	7836.73	7.09 ± 0.03	7.17 ± 0.05	7.20 ± 0.04	–	–
		8068.99	7.04 ± 0.04	7.15 ± 0.04	7.15 ± 0.05	–	–
7963	λ Cyg	8068.98	4.73 ± 0.03	4.78 ± 0.03	4.80 ± 0.03	4.85 ± 0.08	–
7983		7641.96	6.50 ± 0.03	6.53 ± 0.04	6.51 ± 0.04	6.41 ± 0.04	6.21 ± 0.18
		8060.07	6.35 ± 0.03	6.32 ± 0.03	6.21 ± 0.03	–	–
8009		7836.76	6.75 ± 0.08	6.77 ± 0.08	–	–	–
		8067.05	6.76 ± 0.03	6.77 ± 0.04	6.68 ± 0.04	–	–
8047	59 Cyg	7803.72	4.63 ± 0.03	4.54 ± 0.03	4.30 ± 0.04	3.96 ± 0.06	–
		7879.74	4.60 ± 0.04	4.46 ± 0.05	4.28 ± 0.06	3.96 ± 0.07	–
		8059.09	4.59 ± 0.04	4.49 ± 0.06	4.29 ± 0.05	3.99 ± 0.11	–
8053	60 Cyg	7803.69	5.89 ± 0.03	5.96 ± 0.03	5.97 ± 0.03	6.02 ± 0.19	–
		8060.05	5.90 ± 0.03	6.00 ± 0.03	5.98 ± 0.03	6.07 ± 0.07	–
8103		7817.83	6.94 ± 0.03	6.98 ± 0.04	6.98 ± 0.04	–	–
		8067.07	6.99 ± 0.03	7.08 ± 0.04	7.01 ± 0.05	–	–
8146	υ Cyg	7836.81	4.33 ± 0.04	4.25 ± 0.05	4.19 ± 0.05	–	–
		8060.06	4.40 ± 0.03	4.32 ± 0.03	4.09 ± 0.03	3.72 ± 0.06	–
8153		7836.87	5.98 ± 0.04	5.92 ± 0.05	5.87 ± 0.05	–	–
		8069.04	5.94 ± 0.03	5.83 ± 0.04	5.72 ± 0.04	–	–
8171	6 Cep	7836.89	5.25 ± 0.04	5.24 ± 0.05	5.19 ± 0.05	–	–
		8059.12	5.13 ± 0.04	5.18 ± 0.07	5.13 ± 0.05	4.88 ± 0.12	–
8259		7836.83	6.12 ± 0.04	6.13 ± 0.05	6.09 ± 0.05	–	–
		8069.05	6.16 ± 0.03	6.14 ± 0.03	6.10 ± 0.03	–	–
8260	ε Cap	8059.13	7.31 ± 0.06	7.42 ± 0.10	7.38 ± 0.09	7.07 ± 0.13	–
8356	16 Peg	8069.06	5.51 ± 0.03	5.56 ± 0.03	5.57 ± 0.04	5.59 ± 0.08	–
8375		7836.91	6.14 ± 0.04	6.22 ± 0.05	6.17 ± 0.05	–	–
		8069.07	6.14 ± 0.03	6.16 ± 0.03	6.16 ± 0.03	–	–
8386	η PsA	8067.09	5.55 ± 0.03	5.64 ± 0.04	5.61 ± 0.04	5.12 ± 0.17	–
		8069.09	5.62 ± 0.03	5.64 ± 0.04	5.60 ± 0.04	5.45 ± 0.10	–
8402	o Aqr	8067.10	4.83 ± 0.03	4.85 ± 0.04	4.85 ± 0.04	4.66 ± 0.07	–
8408		8069.10	6.59 ± 0.03	6.71 ± 0.04	6.62 ± 0.04	–	–
8438	25 Peg	8069.08	5.93 ± 0.03	5.98 ± 0.03	5.99 ± 0.03	6.17 ± 0.10	–
8520	31 Peg	8060.08	5.13 ± 0.03	5.10 ± 0.03	4.95 ± 0.04	4.60 ± 0.05	–
8539	π Aqr	8060.09	5.16 ± 0.03	5.19 ± 0.03	5.04 ± 0.03	4.62 ± 0.07	–
8603	8 Lac	8070.13	5.99 ± 0.03	6.00 ± 0.03	5.92 ± 0.03	–	–
8628	ε PsA	8069.11	4.57 ± 0.03	4.69 ± 0.05	4.79 ± 0.05	4.46 ± 0.07	–
8682		8070.09	6.31 ± 0.03	6.38 ± 0.04	6.37 ± 0.04	–	–
8690	14 Lac	8070.10	5.57 ± 0.03	5.45 ± 0.03	5.35 ± 0.03	–	–
8731	EW Lac	7642.01	5.36 ± 0.03	5.37 ± 0.04	5.24 ± 0.04	5.00 ± 0.04	4.79 ± 0.10
		7797.73	5.40 ± 0.03	5.40 ± 0.03	5.30 ± 0.03	–	–
		7879.77	5.45 ± 0.04	5.43 ± 0.05	5.30 ± 0.05	5.07 ± 0.07	–
		8060.11	5.47 ± 0.03	5.38 ± 0.03	5.25 ± 0.03	4.96 ± 0.06	–
		8246.66	5.32 ± 0.04	5.21 ± 0.05	5.08 ± 0.05	5.18 ± 0.10	–
8758		7817.88	6.78 ± 0.03	6.81 ± 0.04	6.78 ± 0.04	–	–
		8070.11	6.74 ± 0.03	6.72 ± 0.04	6.65 ± 0.04	–	–
8762	o And	7806.86	3.81 ± 0.03	3.84 ± 0.03	3.85 ± 0.03	3.89 ± 0.04	–
		8060.12	3.79 ± 0.03	3.82 ± 0.03	3.84 ± 0.03	4.11 ± 0.04	–
8773	β Psc	8067.12	4.60 ± 0.03	4.59 ± 0.04	4.49 ± 0.04	4.15 ± 0.07	–
9068		7817.92	6.75 ± 0.03	6.84 ± 0.04	6.81 ± 0.04	–	–
		8070.11	6.76 ± 0.03	6.83 ± 0.03	6.83 ± 0.03	–	–
9070	LQ And	8070.12	6.70 ± 0.03	6.77 ± 0.04	6.71 ± 0.04	–	–

Notes to TABLE 1

(a) Observations obtained at MKO are in the periods JD 2447873 to 2447881 and 2448056 to 2448071; and at NOAO Kitt Peak in the period JD 2447639 to 2447643. The remainder of the observations were obtained at RAO.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 2. Mean magnitudes of multi-epoch near-IR observations of Be stars.

HR	Star Name	J	H	K	L	M
7	10 Cas	5.58 ± 0.02 (2)	5.56 ± 0.02 (2)	5.54 ± 0.02 (2)	–	–
193	o Cas	4.36 ± 0.02 (2)	4.27 ± 0.03 (2)	4.09 ± 0.03 (2)	3.84 ± 0.07 (2)	–
264	γ cas	2.17 ± 0.02 (2)	2.06 ± 0.03 (2)	1.87 ± 0.03 (2)	1.50 ± 0.04 (2)	1.24 ± 0.24
335	ϕ And	4.37 ± 0.02 (2)	4.37 ± 0.03 (2)	4.35 ± 0.03 (2)	4.24 ± 0.05	–
496	ϕ Per	3.93 ± 0.04	3.82 ± 0.05	3.55 ± 0.05	3.09 ± 0.07	–
894		6.24 ± 0.03	6.24 ± 0.04	6.19 ± 0.04	–	–
985	BK Cam	4.93 ± 0.02 (2)	4.88 ± 0.03 (2)	4.72 ± 0.03 (2)	4.41 ± 0.04 (2)	–
1087	ψ Per	4.38 ± 0.02 (2)	4.34 ± 0.03 (2)	4.20 ± 0.03 (2)	3.85 ± 0.04 (2)	–
1126	13 Tau	5.81 ± 0.02 (2)	5.83 ± 0.02 (2)	5.82 ± 0.02 (2)	5.61 ± 0.06 (2)	–
1142	17 Tau	3.96 ± 0.02 (2)	4.02 ± 0.02 (2)	4.01 ± 0.03 (2)	3.91 ± 0.04	3.99 ± 0.30
1156	23 Tau	4.25 ± 0.02 (2)	4.22 ± 0.03 (2)	4.11 ± 0.03 (2)	3.77 ± 0.05	3.42 ± 0.19
1160		6.00 ± 0.04	5.94 ± 0.06	6.05 ± 0.07	–	–
1165	η Tau	3.08 ± 0.02 (3)	3.11 ± 0.02 (3)	3.09 ± 0.02 (3)	2.97 ± 0.04 (3)	3.06 ± 0.21
1180	28 Tau	5.16 ± 0.02 (2)	5.14 ± 0.02 (2)	5.05 ± 0.03 (2)	4.68 ± 0.05	3.83 ± 0.24
1273	48 Per	3.96 ± 0.02 (2)	3.94 ± 0.03 (2)	3.81 ± 0.03 (2)	3.56 ± 0.04 (2)	–
1500		5.90 ± 0.02 (2)	5.87 ± 0.03 (2)	5.82 ± 0.03 (2)	5.59 ± 0.07	–
1622	11 Cam	5.22 ± 0.04	5.13 ± 0.05	4.99 ± 0.05	4.59 ± 0.06	–
1660	105 Tau	5.48 ± 0.02 (2)	5.30 ± 0.03 (2)	5.08 ± 0.03 (2)	4.76 ± 0.06 (2)	–
1789	ψ Ori	5.13 ± 0.03	5.22 ± 0.03	5.27 ± 0.04	5.28 ± 0.05	–
1858	120 Tau	5.74 ± 0.02 (2)	5.69 ± 0.02 (2)	5.60 ± 0.03 (2)	5.21 ± 0.08	–
1910	ζ Tau	3.32 ± 0.02 (2)	3.29 ± 0.03 (2)	3.12 ± 0.03 (2)	2.79 ± 0.05 (2)	–
1934	ω Ori	4.48 ± 0.02 (2)	4.40 ± 0.03 (2)	4.26 ± 0.03 (2)	3.74 ± 0.05 (2)	3.60 ± 0.22
2309		6.28 ± 0.03	6.35 ± 0.04	6.34 ± 0.04	6.28 ± 0.11	–
2343	ν Gem	4.40 ± 0.03	4.45 ± 0.04	4.43 ± 0.05	4.37 ± 0.05	–
2356	β Mon	4.29 ± 0.04	4.14 ± 0.05	3.98 ± 0.05	3.90 ± 0.14	–
2568	ψ^9 Aur	6.02 ± 0.03	6.03 ± 0.05	5.97 ± 0.05	5.82 ± 0.07	–
2825	FW CMa	5.25 ± 0.03	5.19 ± 0.04	5.04 ± 0.04	4.71 ± 0.05	–
2845	β CMi	3.06 ± 0.02 (3)	3.09 ± 0.02 (3)	3.06 ± 0.02 (3)	3.00 ± 0.03 (3)	2.82 ± 0.15
3135		6.49 ± 0.04	6.39 ± 0.05	6.19 ± 0.05	5.82 ± 0.06	5.72 ± 0.19
3858		5.17 ± 0.04	5.06 ± 0.05	4.89 ± 0.05	4.69 ± 0.05	4.35 ± 0.12
3946		6.18 ± 0.04	6.10 ± 0.05	5.93 ± 0.05	5.68 ± 0.05	5.54 ± 0.20
4123		5.60 ± 0.03 (2)	5.62 ± 0.03 (2)	5.56 ± 0.04 (2)	5.54 ± 0.04 (2)	5.48 ± 0.18
4696	ζ Crv	5.45 ± 0.02 (2)	5.47 ± 0.03 (2)	5.44 ± 0.04 (2)	5.37 ± 0.04 (2)	5.60 ± 0.24
4787	κ Dra	4.02 ± 0.02 (4)	4.01 ± 0.02 (4)	3.89 ± 0.02 (4)	3.59 ± 0.02 (4)	3.31 ± 0.07 (3)
5250	47 Hya	5.39 ± 0.03	5.47 ± 0.03	5.45 ± 0.03	5.50 ± 0.08	–
5778	θ CrB	4.42 ± 0.02 (2)	4.48 ± 0.03 (2)	4.48 ± 0.03 (2)	4.42 ± 0.04 (2)	4.52 ± 0.16
5907		5.46 ± 0.03	5.46 ± 0.03	5.45 ± 0.03	5.45 ± 0.07	–
5938	4 Her	5.96 ± 0.01 (5)	5.98 ± 0.02 (5)	5.99 ± 0.02 (5)	6.01 ± 0.03 (3)	5.96 ± 0.10 (3)
5941	48 Lib	4.91 ± 0.03 (2)	4.86 ± 0.04 (2)	4.72 ± 0.03 (2)	4.60 ± 0.04 (2)	4.30 ± 0.10
6118	χ Oph	3.41 ± 0.03 (2)	3.19 ± 0.03 (2)	2.93 ± 0.03 (2)	2.53 ± 0.04 (2)	2.21 ± 0.09 (2)
6422		5.97 ± 0.02 (2)	5.94 ± 0.03 (2)	5.90 ± 0.03 (2)	–	–
6519	51 Oph	4.64 ± 0.04	4.59 ± 0.07	4.32 ± 0.06	3.16 ± 0.14	–
6621	V3894 Sgr	5.92 ± 0.03	5.85 ± 0.04	5.71 ± 0.04	–	–
6664	88 Her	6.97 ± 0.03	6.99 ± 0.04	7.30 ± 0.11	–	–
6712	66 Oph	4.89 ± 0.02 (3)	4.86 ± 0.02 (3)	4.64 ± 0.02 (3)	4.14 ± 0.03 (3)	3.73 ± 0.10
6720		6.45 ± 0.02 (2)	6.46 ± 0.03 (2)	6.40 ± 0.03 (2)	6.23 ± 0.04	5.62 ± 0.29
6873		6.24 ± 0.02 (2)	6.28 ± 0.02 (2)	6.24 ± 0.02 (2)	6.25 ± 0.04	–
6881		5.65 ± 0.03	5.90 ± 0.09	5.65 ± 0.04	–	–
6929	V4031 Sgr	6.47 ± 0.02 (2)	6.47 ± 0.03 (2)	6.46 ± 0.03 (2)	–	–
6971		6.67 ± 0.02 (2)	6.68 ± 0.02 (2)	6.64 ± 0.03 (2)	–	–
6984		6.34 ± 0.02 (2)	6.39 ± 0.02 (2)	6.37 ± 0.02 (2)	6.41 ± 0.25	–
7040	4 Aql	5.11 ± 0.03	5.15 ± 0.02	5.12 ± 0.03	4.96 ± 0.06	–
7084	CX Dra	5.83 ± 0.02 (5)	5.67 ± 0.02 (5)	5.57 ± 0.02 (5)	5.43 ± 0.03 (4)	5.25 ± 0.10 (3)
7158	64 Ser	5.47 ± 0.03	5.47 ± 0.04	5.46 ± 0.04	5.50 ± 0.08	–
7249	V4024 Sgr	5.47 ± 0.03 (2)	5.50 ± 0.04 (2)	5.48 ± 0.04 (2)	5.12 ± 0.05 (2)	4.86 ± 0.12 (2)
7335		7.04 ± 0.03	7.18 ± 0.04	7.22 ± 0.04	–	–
7403		6.35 ± 0.02 (2)	6.33 ± 0.03 (2)	6.15 ± 0.03 (2)	5.84 ± 0.04 (2)	5.86 ± 0.16
7415	V923 Aql	5.86 ± 0.03 (2)	5.84 ± 0.03 (2)	5.84 ± 0.03 (2)	–	–
7418	6 β^2 Cyg	5.24 ± 0.03	5.24 ± 0.03	5.23 ± 0.03	5.15 ± 0.07	–

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 2. (continued)

HR	Star Name	J	H	K	L	M
7457	11 Cyg	6.13 ± 0.02 (2)	6.16 ± 0.02 (2)	6.13 ± 0.03 (2)	5.73 ± 0.22	–
7554	V1339 Aql	6.51 ± 0.03	6.50 ± 0.04	6.34 ± 0.05	–	–
7565	12 Vul	5.01 ± 0.02 (2)	5.04 ± 0.02 (2)	4.98 ± 0.03 (2)	4.77 ± 0.04 (2)	4.76 ± 0.11
7647	25 Cyg	5.35 ± 0.02 (3)	5.39 ± 0.02 (3)	5.31 ± 0.02 (3)	5.13 ± 0.04 (3)	5.02 ± 0.15
7708	28 Cyg	5.17 ± 0.02 (3)	5.17 ± 0.02 (3)	5.08 ± 0.02 (3)	4.90 ± 0.04 (2)	4.81 ± 0.11
7719	20 Vul	6.08 ± 0.02 (2)	6.11 ± 0.02 (2)	6.08 ± 0.02 (2)	5.97 ± 0.04	5.93 ± 0.17
7739		5.18 ± 0.03	5.26 ± 0.03	5.25 ± 0.03	5.21 ± 0.06	–
7789	25 Vul	5.64 ± 0.02 (2)	5.71 ± 0.02 (2)	5.67 ± 0.02 (2)	5.57 ± 0.04 (2)	5.54 ± 0.13
7803		6.19 ± 0.03	6.22 ± 0.05	6.15 ± 0.04	–	–
7807		6.23 ± 0.02 (2)	6.22 ± 0.03 (2)	6.08 ± 0.03 (2)	5.72 ± 0.04 (2)	5.44 ± 0.14
7836	1 Del	6.06 ± 0.03	6.08 ± 0.03	6.18 ± 0.06	–	–
7843		6.00 ± 0.03	5.96 ± 0.04	5.93 ± 0.04	–	–
7890		6.38 ± 0.03	6.40 ± 0.04	6.39 ± 0.04	–	–
7927	V568 Cyg	7.07 ± 0.02 (2)	7.16 ± 0.03 (2)	7.17 ± 0.03 (2)	–	–
7963	λ Cyg	4.73 ± 0.03	4.78 ± 0.03	4.80 ± 0.03	4.85 ± 0.08	–
7983		6.43 ± 0.02 (2)	6.40 ± 0.03 (2)	6.34 ± 0.03 (2)	6.41 ± 0.04	6.21 ± 0.18
8009		6.75 ± 0.03 (2)	6.77 ± 0.04 (2)	6.68 ± 0.04 (2)	–	–
8047	59 Cyg	4.61 ± 0.02 (3)	4.51 ± 0.03 (3)	4.29 ± 0.03 (3)	3.97 ± 0.04 (3)	–
8053	60 Cyg	5.90 ± 0.02 (2)	5.98 ± 0.02 (2)	5.98 ± 0.02 (2)	6.06 ± 0.06 (2)	–
8103		6.96 ± 0.02 (2)	7.02 ± 0.03 (2)	6.99 ± 0.03 (2)	–	–
8146	υ Cyg	4.37 ± 0.02 (2)	4.30 ± 0.03 (2)	4.12 ± 0.03 (2)	3.72 ± 0.06	–
8153		5.95 ± 0.02 (2)	5.86 ± 0.03 (2)	5.78 ± 0.03 (2)	–	–
8171	6 Cep	5.20 ± 0.03 (2)	5.22 ± 0.04 (2)	5.17 ± 0.04 (2)	4.88 ± 0.12	–
8259		6.14 ± 0.02 (2)	6.13 ± 0.03 (2)	6.09 ± 0.03 (2)	–	–
8260	ε Cap	7.31 ± 0.06	7.42 ± 0.10	7.38 ± 0.09	7.07 ± 0.13	–
8356	16 Peg	5.51 ± 0.03	5.56 ± 0.03	5.57 ± 0.04	5.59 ± 0.08	–
8375		6.14 ± 0.02 (2)	6.17 ± 0.03 (2)	6.16 ± 0.03 (2)	–	–
8386	η PsA	5.59 ± 0.02 (2)	5.64 ± 0.03 (2)	5.60 ± 0.03 (2)	5.36 ± 0.09 (2)	–
8402	ο Aqr	4.83 ± 0.03	4.85 ± 0.04	4.85 ± 0.04	4.66 ± 0.07	–
8408		6.59 ± 0.03	6.71 ± 0.04	6.62 ± 0.04	–	–
8438	25 Peg	5.93 ± 0.03	5.98 ± 0.03	5.99 ± 0.03	6.17 ± 0.10	–
8520	31 Peg	5.13 ± 0.03	5.10 ± 0.03	4.95 ± 0.04	4.60 ± 0.05	–
8539	π Aqr	5.16 ± 0.03	5.19 ± 0.03	5.04 ± 0.03	4.62 ± 0.07	–
8603	8 Lac	5.99 ± 0.03	6.00 ± 0.03	5.92 ± 0.03	–	–
8628	ε PsA	4.57 ± 0.03	4.69 ± 0.05	4.79 ± 0.05	4.46 ± 0.07	–
8682		6.31 ± 0.03	6.38 ± 0.04	6.37 ± 0.04	–	–
8690	14 Lac	5.57 ± 0.03	5.45 ± 0.03	5.35 ± 0.03	–	–
8731	EW Lac	5.40 ± 0.01 (5)	5.36 ± 0.02 (5)	5.24 ± 0.02 (5)	5.03 ± 0.03 (4)	4.79 ± 0.10
8758		6.76 ± 0.02 (2)	6.77 ± 0.03 (2)	6.72 ± 0.03 (2)	–	–
8762	ο And	3.80 ± 0.02 (2)	3.83 ± 0.02 (2)	3.84 ± 0.02 (2)	4.00 ± 0.03 (2)	–
8773	β Psc	4.60 ± 0.03	4.59 ± 0.04	4.49 ± 0.04	4.15 ± 0.07	–
9068		6.75 ± 0.02 (2)	6.83 ± 0.02 (2)	6.82 ± 0.02 (2)	–	–
9070	LQ And	6.70 ± 0.03	6.77 ± 0.04	6.71 ± 0.04	–	–

we can calculate the spectral index of the excess spectrum from the excess colors. Z_V can be established from model atmosphere fits to ultraviolet and visual photometry (e.g., see Waters 1986). However, if we assume that $V^0 = V^*$ then $Z_V = 1$. Using this assumption will overestimate the value of α_e if there is an excess at V. This effect is small relative to the uncertainty in the spectral index values. Clearly, in many stars, particularly late-type Be stars where the near-IR color excess is small or zero, it is unlikely that there is any excess in the Paschen continuum. For early type Be stars it is known that there is an excess in the Paschen continuum and the assumption no longer holds. Since γ Cas has the largest excess at J we shall examine this star as a test case. For γ Cas, Waters *et al.* (1991) estimate $Z_{518\ \mathrm{nm}} = 1.363$ based on model atmosphere fits to UV data from the *S* 2/68 satellite (Maceau-Hercot *et al.* 1978). Using the excess colors in Table 2, we find that $\alpha_e - \alpha_*$ is higher than the value obtained for $Z_V = 1$, by 20% for J to K and 10% for K to L. These differences are small compared to the uncertainties in the values of spectral index.

5. NATURE OF THE NEAR-IR EXCESS

5.1 Color–Color Diagrams

The CE($V - m_\lambda$) colors defined in Eq. (1) were calculated for each of the program stars. The excess colors are given in Table 3. We have denoted stars as having an IR excess if the CE($V - m_\lambda$) color differs from zero by more than 4σ. The number of stars that satisfy this criterion at each of the passbands is listed in Table 4. There are few observations at M compared to the other bands. To avoid introducing a selection effect, arising from the fact that the limiting magni-

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 3. Excess colors of Be stars.

HR	Sp Type	CE(V-J)	CE(V-H)	CE(V-K)	CE(V-L)	CE(V-M)	$\alpha_{[J-K]}$	$\alpha_{[K-L]}$	$\alpha_{[12-25]}$
7	B9IIIe	0.01	0.03	0.05					
193	B5IIIe	0.33	0.44	0.63	0.87		0.30 ± 0.16	0.89 ± 0.14	0.78 ± 0.04
264	B0IVe	0.64	0.82	1.03	1.41	1.72	0.68 ± 0.08	0.78 ± 0.35	1.00 ± 0.02
335	B7Ve	−0.01	0.00	0.03	0.15				0.27 ± 0.21
496	B2Vep	0.22	0.36	0.63	1.08		−0.37 ± 0.65	0.23 ± 0.24	0.84 ± 0.02
894	B8Vne	−0.03	−0.02	0.03					
985	B2.5Ven	0.25	0.37	0.54	0.87		0.15 ± 0.27	0.53 ± 0.24	0.46 ± 0.09
1087	B5Ve	−0.03	0.03	0.18	0.53			−0.76 ± 0.80	0.65 ± 0.03
1126	B9Vne	−0.14	−0.16	−0.16	0.04				
1142	B6IIIe	−0.03	−0.04	−0.02	0.08	0.03			
1156	B6IVe	0.04	0.10	0.21	0.55	0.93		−0.46 ± 0.72	
1160	B8Vne	−0.04	−0.01	−0.14					
1165	B7IIIe	−0.02	−0.04	−0.00	0.13	0.05			
1180	B8Vpe	0.07	0.11	0.21	0.58	1.44	−0.32 ± 1.21	−0.68 ± 0.73	
1273	B3Ve	0.15	0.20	0.32	0.57		0.30 ± 0.51	0.41 ± 0.30	0.53 ± 0.04
1500	B8IVne	0.02	0.03	0.06	0.28				
1622	B2.5Ve	0.04	0.17	0.31	0.72			−0.28 ± 0.57	
1660	B2Ve	−0.03	0.09	0.26	0.54			0.07 ± 0.59	0.34 ± 0.02
1789	B1Vpe	0.31	0.29	0.27	0.27		2.08 ± 0.07	1.79 ± 0.01	1.15 ± 0.02
1858	B2IV-Ve	−0.08	−0.03	0.05	0.43				
1910	B4IIIpe	0.10	0.21	0.41	0.76		−1.00 ± 0.98	0.19 ± 0.26	0.90 ± 0.03
1934	B3IIIe	0.34	0.47	0.63	1.15	1.35	0.43 ± 0.17	0.04 ± 0.19	
2309	B5Ve	0.01	−0.02	−0.01	0.04				
2343	B6IIIe	0.03	0.02	0.07	0.13				0.34 ± 0.16
2356	B3Ve	0.54	0.74	0.90	0.99		0.54 ± 0.15	1.54 ± 0.07	1.09 ± 0.04
2568	B8IIIe	−0.03	−0.02	0.03	0.19				
2825	B2.5IVe	0.19	0.28	0.43	0.76		0.14 ± 0.49	0.34 ± 0.23	
2845	B8Ve	0.01	0.00	0.03	0.11	0.29			0.40 ± 0.16
3135	B2.5Ve	0.18	0.32	0.52	0.88	1.04	−0.38 ± 0.76	0.37 ± 0.22	
3858	B6Ve	−0.14	0.01	0.19	0.40	0.77		0.08 ± 0.72	
3946	B4Ve	0.25	0.37	0.56	0.82	0.99	0.08 ± 0.43	0.77 ± 0.15	
4123	B9Vne	0.03	0.02	0.08	0.09	0.16			
4696	B8Vne	−0.05	−0.05	−0.01	0.07	−0.15			
4787	B6IIIpe	0.13	0.19	0.32	0.63	0.93	−0.07 ± 0.43	0.11 ± 0.19	0.78 ± 0.03
5250	B8Vp-shell	−0.05	−0.10	−0.08	−0.12				
5778	B6nne	0.00	−0.02	−0.00	0.07	−0.00			
5907	B2.5Vne	0.05	0.07	0.08	0.08				
5938	B9pe	0.01	0.00	0.01	−0.00	0.07			
5941	B5IIIpe	0.18	0.28	0.42	0.54	0.88	−0.02 ± 0.49	1.17 ± 0.11	1.03 ± 0.04
6118	B2IV:pe	0.37	0.50	0.70	1.04	1.39	0.37 ± 0.19	0.65 ± 0.09	0.84 ± 0.02
6422	B5Vne	0.04	0.01	0.02					
6519	B9.5Ve	0.13	0.17	0.45	1.60		−0.80 ± 1.26	−2.02 ± 1.05	0.71 ± 0.00
6621	B4IVe	0.14	0.19	0.30			0.34 ± 0.60		
6664	Bep-shell	−0.15	−0.16	−0.46					
6712	B2Ve	−0.19	−0.14	0.08	0.57	1.03		−2.60 ± 1.53	0.71 ± 0.02
6720	B8Vne	0.15	0.15	0.21	0.39	1.00	1.01 ± 0.34	0.38 ± 0.45	
6873	B3Ve	−0.01	−0.03	0.00	−0.00				
6881	B8IV-Ve	0.04	−0.22	0.02					
6929	B2IVpe	−0.04	−0.06	−0.07					
6971	B4Ve	0.19	0.22	0.28			0.97 ± 0.30		
6984	B5Vne	−0.01	−0.01	0.02	−0.02				
7040	B9V	0.00	−0.03	0.02	0.17				
7084	B2.5Ve	0.28	0.49	0.59	0.71	0.94	0.18 ± 0.16	1.34 ± 0.04	

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 3. (continued)

HR	Sp Type	CE(V-J)	CE(V-H)	CE(V-K)	CE(V-L)	CE(V-M)	$\alpha_{[J-K]}$	$\alpha_{[K-L]}$	$\alpha_{[12-25]}$
7158	B9IIIep:Hg	0.04	0.02	0.02	−0.02				
7249	B2Ve	0.31	0.33	0.37	0.73	1.04	1.49 ± 0.08	0.06 ± 0.32	
7335	B2IVe	−0.02	−0.09	−0.09					
7403	B3Ve	0.32	0.39	0.59	0.92	0.95	0.42 ± 0.18	0.58 ± 0.10	0.84 ± 0.02
7415	A0e-shell	0.15	0.16	0.15			1.63 ± 0.02		
7418	B8Ve	0.06	0.08	0.10	0.20			0.45 ± 1.46	
7457	B8Vne	0.13	0.13	0.17	0.58		1.15 ± 0.22		
7554	B2.5IVe	0.23	0.28	0.45			0.40 ± 0.44		
7565	B2.5Ve	0.26	0.30	0.36	0.58	0.65	1.09 ± 0.13	0.65 ± 0.19	
7647	B3IVe	0.23	0.26	0.37	0.56	0.73	0.82 ± 0.15	0.74 ± 0.20	
7708	B2.5Ve	0.05	0.11	0.21	0.41	0.55		0.33 ± 0.44	
7719	B7Ve	0.05	0.03	0.09	0.21	0.26		0.06 ± 1.37	
7739	B3V	0.02	0.00	0.05	0.10				
7789	B8IIIne	0.02	−0.03	0.01	0.11	0.15			0.24 ± 0.09
7803	B9V	0.01	−0.02	0.05					
7807	B2Ven	0.12	0.20	0.38	0.75	1.09	0.51 ± 0.72	0.04 ± 0.27	
7836	A1e-shell	0.09	0.07	−0.01					
7843	B9Vne	−0.04	−0.00	0.03					
7890	B7IIIne	0.04	0.04	0.08					
7927	B2IV-Ve	−0.05	−0.08	−0.07					
7963	B5Ve	0.04	0.03	0.02	−0.02				
7983	B4Ve	0.05	0.11	0.18	0.12	0.34			
8009	B8Vnne	0.06	0.07	0.15					
8047	B1ne	0.27	0.40	0.62	0.93		0.06 ± 0.24	0.64 ± 0.11	
8053	B1Ve	−0.02	−0.02	0.01	−0.06				
8103	B4IVpe	0.02	0.03	0.08					
8146	B2Vne	0.30	0.42	0.61	1.02		0.28 ± 0.31	0.34 ± 0.18	
8153	B2IIIe	0.00	0.03	0.07					
8171	B3IVe	0.08	0.08	0.13	0.41			−0.77 ± 3.24	
8259	B9IIIe	−0.10	−0.12	−0.08					
8356	B3Ve	−0.04	−0.02	−0.01	−0.01				
8375	B2.5Ve	−0.15	−0.14	−0.13					0.31 ± 0.04
8386	B8 Ve	−0.01	−0.03	0.01	0.26				
8402	B7IVe	−0.05	−0.07	−0.06	0.13				0.21 ± 0.07
8408	B4IVne	−0.32	−0.38	−0.27					
8438	B7Vne	0.03	−0.00	0.01	−0.16				
8520	B2IV-Ve	0.18	0.27	0.43	0.79		0.02 ± 0.49	0.21 ± 0.28	
8539	B1Ve	−0.41	−0.41	−0.27	0.14				
8603	B2Ve	0.08	0.13	0.23					
8628	B8Ve	−0.19	−0.28	−0.37	−0.03				0.37 ± 0.17
8682	B5Vne	−0.04	−0.08	−0.08					
8690	B3IV:e	0.17	0.28	0.35			0.35 ± 0.44		
8731	B4IIIpe	0.28	0.37	0.51	0.72	1.02	0.56 ± 0.13	0.87 ± 0.09	0.90 ± 0.02
8758	B3Vpe	0.03	0.08	0.14					
8762	B6IIIpe+A2p	0.01	0.01	0.01	−0.15				
8773	B6Ve	0.19	0.24	0.35	0.70		0.45 ± 0.42	0.05 ± 0.46	0.16 ± 0.15
9068	B6IVe	−0.00	−0.04	−0.02					
9070	B4Ven	0.04	0.01	0.08					

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

TABLE 4. Number of stars with excess.

Passband	Total	$> 4\sigma$	%
J	101	33	33
H	101	36	36
K	101	43	43
L	72	40	56
M	30	19	63

tude at M is such that we preferentially detect stars with excess, we have excluded our M band observations in the remaining discussion.

It is interesting to compare the number of stars with detectable excess emission in this survey with those detected by *IRAS*. Coté & Waters (1987) include only 48 of the 101 stars in this survey in their list of stars with reliable *IRAS* detections at 12 μm. All 48 stars exhibit a 4σ color excess at 12 μm. The rate of detection of stars with color excess is less at 12 μm than at L or M bands. Near-IR observations provide a more efficient method for detecting excess in Be stars than the *IRAS* survey. This is due to the *IRAS* limiting magnitude of + 5 at 12 μm.

In Fig. 1 the CE($V - m_\lambda$) color excesses of the survey stars at each IR wavelength are plotted against the intrinsic ($B - V$) colors. The error bars are the 1σ uncertainties quoted in Table 2. Previous near-IR surveys of Be stars indicate that only ~20%–25% of Be stars are variable in the near-IR at the level of a few tenths of a magnitude. It is reasonable to assume that any trends indicated by the color–color diagrams are not affected by variations.

There are several features in the color–color diagrams of Fig. 1 that are immediately apparent:

(1) The largest color excesses occur at lower values of intrinsic color.

(2) For a given spectral type the largest excess color occurs at longer wavelengths.

(3) Two stars, γ Cas and β Mon, have large color excess as compared to the majority of stars at the J, H, and K passbands. At L, the IR excess of 51 Oph is dramatically larger than any other late-type star, and is also much larger than its value at K band.

(4) There are a few stars for which the color excess is negative. These include 88 Her, 66 Oph, π Aqr and HR 8408. This effect diminishes at longer wavelengths.

Some of these effects can be easily understood. Figure 2(a) of CE($J - K$) vs CE($K - L$) shows that the majority of Be stars have a larger value of CE($K - L$) than CE($J - K$), indicating an increase in excess, i.e., Z_λ with wavelength. The increase of Z_λ with wavelength continues at least to far-IR wavelengths [Fig. 2(b)] and in a few stars out to millimeter wavelengths (see Waters *et al.* 1991). Far-IR data in Fig. 2(b) are from Waters *et al.* (1987). In the case of excess increasing with wavelength, $CE(m_{\lambda_1} - m_{\lambda_2}) > 0$, which requires that $a > b$. This further implies that $\alpha_e < \alpha_*$ [see Eq. (8)]. At near-IR wavelengths, the photospheres of hot stars behave as $\alpha_* \simeq 2$ and therefore

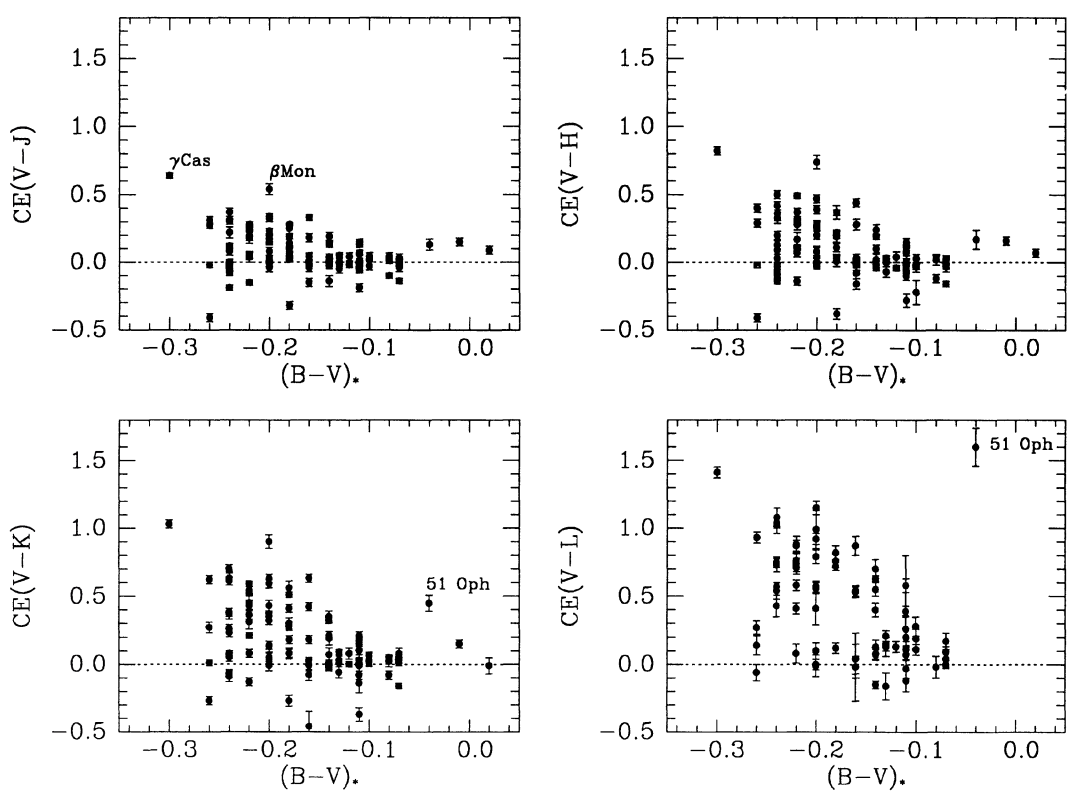

FIG. 1. CE($V - m_\lambda$) colors as a function of intrinsic $(B - V)_*$ color.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

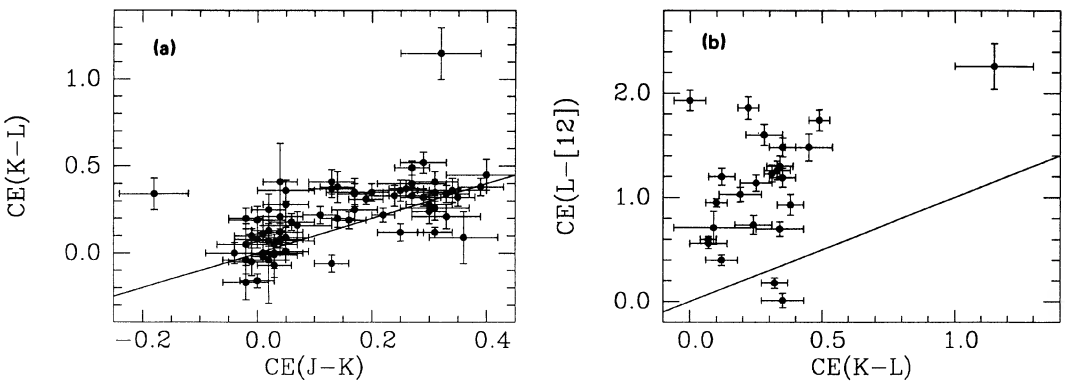

FIG. 2. Near-IR excess color–color diagrams. Solid lines are the locii of points for which the abcissa and ordinate are equal.

for excess increasing with wavelength, $\alpha_e < 2$. This is the case for optically thin or partially optically thin bremsstrahlung. We note that it is not possible to discriminate between free–free emission and emission from very hot dust ($T \sim 3000$ K) using near-IR colors alone (Allen 1973). However, observations in the far-IR of Be stars show that the circumstellar emission is due to free–free and bound–free processes in virtually all cases. 51 Oph is the only star in the survey that is thought to have IR emission arising from hot dust (see Sec. 7).

The largest excesses occur in earlier spectral types. It has already been mentioned that some early type Be stars have an excess of emission in the Paschen continuum. Since excess increases with wavelength, large excess at near-IR wavelengths is expected for some stars, as seen in γ Cas and β Mon.

A negative $(V - m_\lambda)$ color excess can be produced if $Z_V > Z_\lambda$. This requires that $Z_V > 1$ (Recall that we assumed that $Z_V = 1$ in our calculations). A negative color excess can also be explained by variability. The V magnitudes used in the calculation of color excess are from the BSC and were not obtained simultaneously with the IR observations. If V, or f_λ^e, decreased in the period between the V observations quoted in the BSC (Nicolet 1978) and the epoch of the IR observations then it is possible to get negative color excess. The stars 88 Her, 66 Oph and π Aqr have extensive histories of optical variations (see Barylak & Doazon 1985; Slettebak 1982; Dachs *et al.* 1986). The question of IR variability will be addressed in Paper II.

5.2 Correlation of Excess with Stellar Parameters

Of interest is whether there is any correlation between excess colors and stellar parameters such as spectral type and the projected rotational velocity $v \sin i$. As mentioned in the previous section, the largest color excess occurs for stars with smaller values of intrinsic color. This implies that stars of earlier spectral type have larger excess. In Fig. 3 the $(V - L)$ color excess is plotted against spectral type, with luminosity class indicated by different symbols. For a particular spectral type there is a large scatter in the color excess, with no apparent dependence on luminosity class, implying that the color excess is not linked to stellar luminosity. However, there is a well-defined upper envelope to the color excess (excluding 51 Oph) that increases from late to early spectral types. Be stars of a given spectral type do not have an arbitrarily large excess. Most of the late type stars in the survey show little or no near-IR excess color, even at L band (see Fig. 1). Coté and Waters (1987) reached the same conclusions from analysis of *IRAS* far-IR excess colors.

Stellar rotation must play a role in the formation of a Be star from a B star since Be stars typically have large projected rotational velocities compared to normal B stars (Slettebak 1966). Struve (1931) suggested that the formation of the circumstellar envelopes around Be stars is due to the ejection of material induced by rapid rotation. If this is the case then a correlation between excess and rotational velocity is expected. In Fig. 4 the excess at L band is shown as a function of $v \sin i$. Values of $v \sin i$ were taken from Slettebak (1982) and the BSC. No correlation is apparent within the large scatter in the data. Gehrz *et al.* (1974) and Coté & Waters (1987) also found no apparent correlation between $v \sin i$ and near- and far-IR excess, respectively. There is a small number of stars with large color excess that have a lower $v \sin i$ than the majority of stars in the survey. This

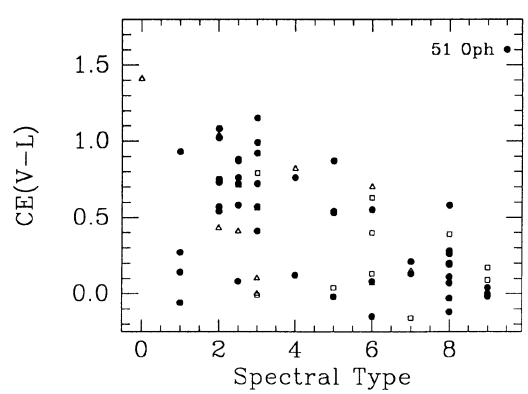

FIG. 3. The excess $(V - L)$ color as a function of spectral type. Luminosity classes are represented by □ (III), △ (IV), and ● (V).

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

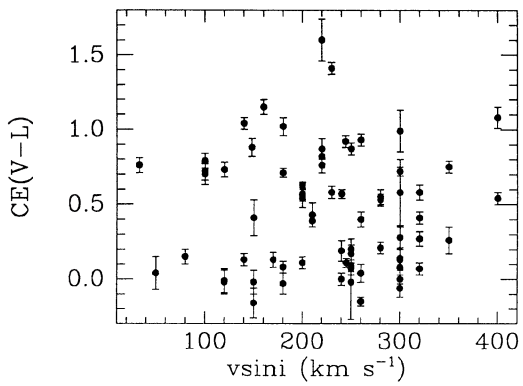

FIG. 4. The excess $(V-L)$ color as a function of $v \sin i$.

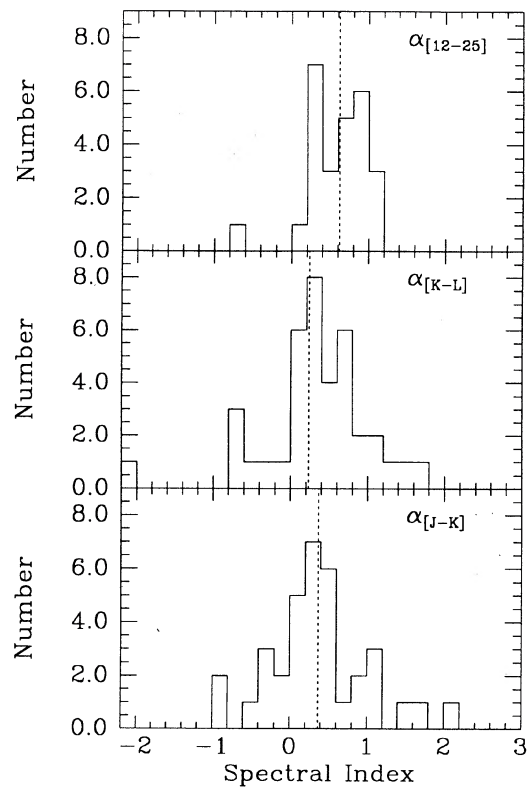

FIG. 5. Distributions of IR spectral index. Vertical dashed lines are the mean values of the distributions.

may be accounted for by the influence of the angle between the line of sight and the rotation axis of the star. If the survey stars have a uniform distribution of inclination angles a small number of stars will have low values of $v \sin i$ (Briot 1986). Clearly, a connection between rotational velocity and the near-IR excess of Be stars is not evident from our observations. However, in a study of the far-IR excess of both B and Be stars, Waters *et al.* (1987) conclude that rotation facilitates the presence of circumstellar material but other mechanism(s) determine whether or not a B star becomes a Be star.

6. THE SHAPE OF THE EXCESS SPECTRUM

In Sec. 4 we discussed how the spectral index of the excess spectrum can be determined from excess colors. Using Eq. (8) we have calculated the spectral indices for the *J–K* and *K–L* passbands ($\alpha_{[J-K]}$ and $\alpha_{[K-L]}$, respectively) for stars that have color excess greater than 4σ. The uncertainties in the individual spectral index values are large. This is to be expected since at shorter wavelengths the excess is typically small and at longer wavelengths, where Z_λ is larger, the uncertainty in the observations increases. While this prevents us from discussing the individual values in any detail (except for a few cases) the collective properties of the spectral index values can be examined.

The distributions of spectral indices $\alpha_{[J-K]}$, $\alpha_{[K-L]}$ and $\alpha_{[12-25]}$ for the survey sample are shown in Fig. 5. The data for the far-IR wavelengths are again taken from Waters *et al.* (1987). The distribution means are indicated by the vertical dashed lines. The spectral indices at the three wavelength regions are very similar. The peak of the $\alpha_{[12-25]}$ appears to be shifted slightly to steeper spectral index. However, the means of the three distributions ($\overline{\alpha}_{[J-K]} = +0.37$; $\overline{\alpha}_{[K-L]} = +0.23$; $\overline{\alpha}_{[12-25]} = +0.62$) are not significantly different. This indicates that, on average, the power law spectral index of the excess emission remains unchanged from the near- to far-IR wavelength regions. For isothermal circumstellar material with an axis-symmetric geometry, constant spectral index implies a constant radial density distribution over the range of radii responsible for the origin of the near- and far-IR radiation.

The spectrum of free–free emission has spectral index from -0.1 for optically thin emission to $+2.0$ for optically thick emission. The data suggest that there are some Be stars that have excess spectra with near-IR spectral indices less than -0.1. Spectral indices lower than -0.1 in the spectra can arise from two possible mechanisms. If the emission arises from hot dust rather than from free–free processes then a large negative spectral index is possible if the near IR is on the shortward side of the hot dust turnover. This is thought to be the case for 51 Oph as described in Sec. 7.

The other possibility is that the circumstellar material around a star is cooler than the photosphere. Then, absorption of photospheric emission occurs and a steep negative spectrum can occur. To see this, note that the total flux from a Be star is made up of two components: the flux from the circumstellar material, f^{cs}, and the attenuated photospheric flux (assuming that some circumstellar material lies along the line of sight to the star). For isothermal circumstellar material of temperature T_{cs} and optical depth τ_{cs} the excess flux due to free–free and bound–free emission from solid angle Ω may be written as

$$f_\lambda^e = \int_{\Omega \neq \Omega_*} B_\lambda(T_{cs})(1-e^{-\tau_{cs}})d\Omega + \int_{\Omega=\Omega_*} [B_\lambda(T_{cs}) - I_\lambda^*](1-e^{-\tau_{cs}})d\Omega, \quad (9)$$

where Ω_* is the solid angle subtended by the stellar photosphere. If we approximate the photosphere as a blackbody, i.e., $I_\lambda^* = B_\lambda(T_*)$, then the second term in Eq. (9) becomes

$$B_\lambda(T_*) \int_{\Omega=\Omega_*} \left(\frac{T_{cs}}{T_*} - 1\right)(1-e^{-\tau_{cs}})d\Omega$$

This term is either negative or zero since $T_{cs} \lesssim T_*$. When

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1991AJ....102.1753D

$\Omega_{cs} \gg \Omega_*$ the first term on the righthand side of Eq. (9) dominates and $f^e \approx f^{cs}$. The lowest spectral index in this case is -0.1. However, if the solid angle of the photosphere is a major fraction of the solid angle subtended by the circumstellar envelope, i.e., $\Omega_{cs} \sim \Omega_*$, or the circumstellar envelope is very cool, then the contribution of the second term becomes more significant and a steep turn down in the excess spectrum may occur. When $\tau_{cs} \gtrsim 1$, the magnitude of this term increases as wavelength decreases for a particular T_{cs}, since $B_\lambda(T_*) \propto \lambda^{-2}$, and the slope of the turndown would decrease.

7. INDIVIDUAL STARS

There are several stars in the survey that exhibit different behavior compared to the bulk of the Be stars in the survey.

The color excess of 51 Oph increases dramatically between K and L band and is much larger than for the other late-type stars in the survey. Further, the far-IR colors of this star are very different from those of many Be stars (Coté & Waters 1987). The values of spectral indices indicate a turnover in the excess emission spectrum from a steep positive spectral index in the near IR to a shallow negative value in the far IR. Waters *et al.* (1988) have shown that these observations indicate the presence of hot dust around the star.

The stars γ Cas and β Mon are also noteworthy in that their color excess values are larger at the shorter wavelength near-IR bands relative to the other stars in the survey. It is known that γ Cas has an excess in the Paschen continuum, and a large excess at near-IR wavelengths is expected. The star β Mon is actually a triple system where the widest separation of the three components is less than the smallest detector aperture used in the survey observations. The observed emission is therefore from the whole system rather than an individual star. Thus, we have to be careful in interpreting the observations of β Mon. It should be noted that both γ Cas and one of the components of β Mon, β Mon A, have been detected at radio wavelengths (Taylor *et al.* 1990). The connection between near-IR excess and radio emission is not clear since several other radio emitting Be stars (ψ Per, β CMi, η Tau, and EW Lac) have a color excess similar to the bulk of the survey stars. Variability further confuses the issue since β Mon A, EW Lac, and possibly γ Cas, have variable radio emission (Dougherty *et al.* 1991).

8. SUMMARY

In this paper we have reported the the observations of 101 bright Be stars between 1 μm and 5 μm. The percentage of stars with detectable color excess was found to increase with wavelength. Color–color diagrams have been used to examine the nature of the $(V - m_\lambda)$ excess. It was found that the color excess increases with wavelength, which is expected for excess emission arising from free–free and bound–free processes. The largest color excess was found to occur for earlier spectral type Be stars. There is no apparent dependence of excess on stellar luminosity. However, there is an upper limit to the color excess for a particular spectral type that increases from late to early types. The existence of an upper limit implies that Be stars of given spectral type do not have arbitrarily large color excess. The excess colors also show no obvious strong correlation with projected rotational velocity.

The values of spectral index for J–K and K–L bands were compared to the far-IR *IRAS* spectral indices. The resulting distributions indicate that, on average, the spectral index is constant throughout the near- and far-IR wavelength regime. This implies that the radial density gradient is constant in the circumstellar material that gives rise to the IR radiation.

We would like to thank the Institute for Astronomy at the University of Hawaii for allowing us to use the MKO 0.6 m telescope. Drs. W. Sinton, A. Pickles, and R. Wainscoat provided valuable support during our MKO observing runs. SMD acknowledges financial support from an Izaak Walton Killam Memorial Scholarship. This research was supported by the Natural Sciences and Engineering Research Council of Canada.

REFERENCES

Allen, D. A. 1973, MNRAS, 161, 145
Ashok, N. M., Bhatt, H. C., Kulkarni, P. V., and Joshi, S. C. 1984, MNRAS, 211, 471
Baade, D., Ed. 1991, Rapid Variability of OB stars: Nature and Diagnostic Value, ESO conference and Workshop Proceedings No. 36
Barylak, M., and Doazon, V. 1985, A&A, 159, 65
Briot, D. 1986, A&A, 163, 67
Coté, J., and Waters, L. B. F. M. 1987, A&A, 176, 93
Dachs, J., Engels, D., and Kiehling, R. 1988, A&A, 194, 167
Dachs, J., Hanuschik, R., Kaiser, D., Ballereau, D., Bouchet, P., Kiehling, R., Kozok, J., Rudolph, R., and Schlosser, W. 1986, A&AS, 63, 87
Dachs, J., and Wamsteker, W. 1982, A&A, 107, 240
Doazon, V. 1982, in The B Stars With and Without Emssion Lines, edited by A. Underhill and V. Doazon (NASA-SP 456)
Doazon, V., Thomas, R. N., and Barylak, M. 1986, A&A, 159, 75
Dougherty, S. M., Taylor, A. R., and Waters, L. B. F. M. 1991, A&A, 248, 175
Fitzgerald, M. P. 1970, A&A, 4, 234
Gehrz, R. D., Hackwell, J. A., and Jones, T. W. 1974, ApJ, 191, 675
Hofleit, D., and Jascheck, C. 1982, The Bright Star Catalogue, Yale University Observatory, New Haven
Koorneef, J. 1983a, A&AS, 51, 489
Koorneef, J. 1983b, A&A, 128, 84
Lamers, H. J. G. L. M., and Waters, L. B. F. M. 1984, A&A, 136, 37
Maceau-Hercot, D., Jamar, C., Monfils, A., Thompson, G. I., Houziaux, L., and Wilson, R. 1978, Supplement to the Ultraviolet Bright-Star Spectrophotometric Catalogue (ESA SR-28)
Nicolet, B. 1978, A&AS, 34, 1
Poeckert, R., and Marlborough, J. M. 1978, ApJS, 38, 229
Schild, R. 1978, ApJS, 37, 77
Slettebak, A. 1966, ApJ, 145, 121
Slettebak, A. 1982, ApJS, 50, 55
Slettebak, A., and Snow, T., Eds. 1987, Physics of Be Stars (Cambridge University Press, Cambridge)
Struve, O. 1931, ApJ, 73, 94
Taylor, A. R., Waters, L. B. F. M., Bjorkman, K. S., and Dougherty, S. M. 1990, A&A, 231, 453
Taylor, A. R., Waters, L. B. F. M., Lamers, H. J. G. L. M., Bjorkman, K. S., and Persei, P. 1987, MNRAS, 228, 811
Waters, L. B. F. M. 1986, A&A, 162, 121
Waters, L. B. F. M., Boland, W., Taylor, A. R., van de Stadt, H., and Lamers, H. J. G. L. M. 1989, A&AL, 213, L19
Waters, L. B. F. M., Coté, J., and Geballe, T. R. 1988, A&A, 203, 348
Waters, L. B. F. M., Coté, J., and Lamers, H. J. G. L. M. 1987, A&A, 185, 206
Waters, L. B. F. M., van der Veen, W. E. C. J., Taylor, A. R., Marlborough, J. M., and Dougherty, S. M. 1991, A&A, 244, 120
Woolf, N. J., Stein, W. A., and Strittmatter, P. A. 1970, A&A, 9, 252

© American Astronomical Society • Provided by the NASA Astrophysics Data System